

A Logistic Regression Approach for Dengue Prediction Using Clinical Indicators

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Abstract—Machine learning techniques are widely used in healthcare to assist in disease prediction and diagnosis. This study applies logistic regression to the Dengue Disease Dataset from Mendeley Data, which contains clinical and laboratory measurements such as Age, Sex, Haemoglobin, WBC Count, Differential Count, RBC Panel, Platelet Count, and PDW. The main objective of the study is to predict whether a patient is infected with dengue based on these medical indicators. Several experiments were conducted to evaluate the effects of hyperparameter tuning, feature selection, and decision threshold adjustment on model performance. The results show that WBC Count and Platelet Count are the most important predictors for dengue classification. The best-performing configuration used all available features with $C = 1.0$ and a decision threshold of 0.5, achieving an accuracy of 97.47% with strong precision, recall, and F1-score values. These findings demonstrate that logistic regression is an effective approach for dengue prediction using clinical data.

Keywords—Logistic Regression, Dengue Prediction, Machine Learning in Healthcare, Feature Selection, Binary Classification

I. INTRODUCTION

Machine learning techniques are increasingly used in healthcare applications to assist in disease prediction and diagnosis. Among these techniques, logistic regression is one of the most widely used algorithms for binary classification due to its simplicity, interpretability, and effectiveness in modeling probability outcomes [1]. In this activity, a dataset called “dengue disease dataset” from [2] is used for logistic regression to classify whether a patient is likely to have dengue based on medical indicators such as platelet count and other clinical measurements.

Regression models are divided into two types: linear regression and logistic regression, which are used for different types of prediction. Linear Regression predicts continuous numerical values by modeling the relationship of independent variables and dependent variable using a linear equation. On the other hand, logistic regression is designed for categorical outcomes, where the outcome variable is represented into categories such as 0 or 1, yes or no, or disease and non-disease [3].

To convert the linear output of the regression model into a probability value between 0 and 1, logistic regression uses the sigmoid function, also known as logistic function. This function transforms any input into a value within the range of 0 to 1, which enables the model to represent probability for classification [4]. Furthermore, logistic regression is based on the concept log-odds, also known as the logit function. It transforms probabilities into log-odds, allowing a linear relationship between predictors and the outcome variable.

Therefore, the objective of this activity is to apply logistic regression to the Dengue Disease Dataset to classify whether a patient is likely to have dengue based on medical indicators. The activity also explores how model performance can be improved through hyperparameter tuning, feature selection, and decision threshold adjustment, evaluated using metrics such as accuracy, precision, recall, and F1-score.

II. DATA COLLECTION

This section describes the process used to build the logistic regression model for predicting dengue cases. The methodology includes dataset description, data preparation, feature selection, and train-test splitting.

A. Dataset Description

The dataset used in this activity is the “dengue disease dataset” from the Mendeley Data repository [2]. This dataset contains medical records and laboratory measurements that are used to analyze and predict whether a patient is likely to have dengue. The dataset consists of 1003 data rows and 9 column variables, which represent patient demographic information and clinical test results related to dengue diagnosis.

Dataset exploration was conducted as a first step to understand the structure, variables, and data types. The dataset includes attributes such as Age, Sex, Haemoglobin, WBC Count, Differential Count, RBC Panel, Platelet Count, PDW, and Final Output. Most of the variables represent numerical laboratory measurements, while the Final Output serves as the binary target variable indicating whether a patient is diagnosed with dengue or not. Some records also contain missing or inconsistent values, which reflect real-world medical data conditions. By understanding this dataset structure, it helps determine the appropriate data cleaning and preprocessing techniques that were applied before building the logistic regression model.

B. Data Preparation

To address the identified data quality issues, a structured data cleaning and preparation process was performed using the pandas library in Python. The dataset was first examined for missing values, inconsistent entries, and potential duplicates. Unnecessary spaces in column names were removed to avoid errors during analysis. Records with missing values in the target variable Final Output were removed to maintain the integrity of the classification.

Moreover, missing values in numerical attributes such as WBC Count, Platelet Count, and PDW were handled using a median, where missing entries were replaced with the median value of the respective column. This approach helps reduce the influence of extreme values while maintaining the overall distribution of the data. Categorical variables such as Sex were converted into numerical format to make them compatible with the machine learning model.

To improve model performance and ensure consistency among variables, numerical features were normalized using Min-Max scaling, which transforms feature values into a range between 0 and 1. This scaling step helps prevent features with larger numerical ranges from dominating the learning process.

The dataset was also examined for duplicate entries and inconsistencies that might introduce bias into the model training. Any unnecessary or redundant records were removed. These data preparation steps ensured that the dataset was clean, structured, and suitable for training the logistic regression model and conducting further experimental analysis.

C. Feature Selection

Feature selection was performed to identify the relevant variables used to train the logistic regression model. The independent variables, which represent the medical and demographic indicators that may influence dengue diagnosis. These features include Age, Sex, Haemoglobin, WBC Count, Differential Count, RBC Panel, Platelet Count, and PDW, which provide important clinical information about the patient's health condition.

The target variable (Y) used for classification is the Final Output, which represents a binary outcome indicating whether a patient is diagnosed with dengue or not. This variable serves as the label that the logistic regression model aims to predict based on the selected input features.

D. Train-Test Split

To evaluate the performance of the logistic regression model, the dataset was divided into training and testing sets. The training set was used to train the logistic regression model and allow it to learn the relationship between the input features and the target variable. The testing set was used to evaluate the model's performance on unseen data and determine how well the model performed. In this activity, the dataset was split using an 80:20 ratio, where 80% of the data was allocated for training and 20% was reserved for testing. This approach ensures that the model is trained on a sufficient amount of data while still maintaining a separate portion for unbiased evaluation of the model's predictive performance.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Hyperparameter Tuning Experiment

Changing the C value in logistic regression affected both the model complexity and prediction accuracy because C controls the strength of regularization.

C Tuning Table Evaluation				
C	Accuracy	Precision	Recall	F1
0.001	67.68%	0.46	0.68	0.55
1.0	96.97%	0.97	0.97	0.97
1000	98.48%	0.98	0.98	0.98

Table. 1. C Tuning Table Evaluation

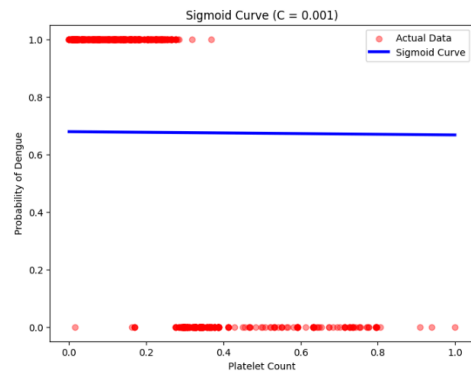


Fig.1 C = 0.001 Sigmoid Curve

Based on the table and figure 1, when $C = 0.001$, it shows strong regulation which can result in underfitting and forces the model to become simple. This strong regulation resulted in a very flat sigmoid curve, indicating that the model could not clearly separate the two classes. As a result, the model predicted almost as class 1, producing an accuracy of 67.68% and poor classification failing to correctly classify the class 0.

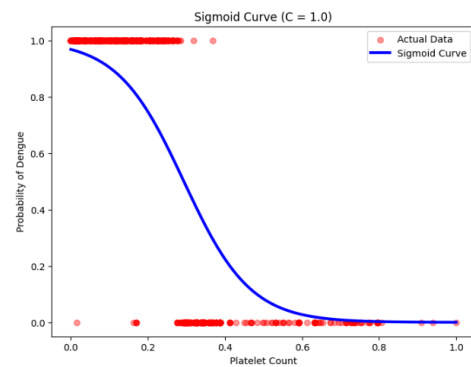


Fig. 2 C = 1.0 Sigmoid Curve

Moreover, figure 2 when $C = 1.0$, the regulation becomes balanced, allowing the model to fit the data more accurately. The sigmoid curve became smoother and better separating the two classes. This improved the model's predictive ability, resulting in higher accuracy of 96.97% and a good classification matrix.

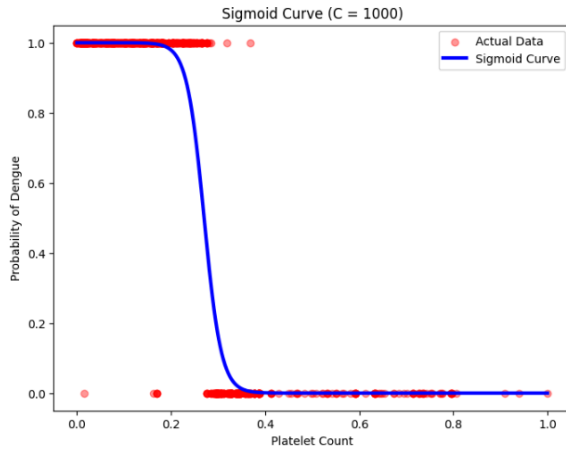


Fig. 3 $C = 1000$ Sigmoid Curve

Finally, for the last experiment figure 3, when $C = 1000$, the regulation becomes weak and maybe prone to overfitting, allowing the model to become flexible. The sigmoid curve for this becomes much steeper, indicating a more strong decision boundary between the two classes. This produced the highest accuracy of 98.48%, as the model was able to correctly classify most data.

B. Feature Engineering and Selection

Feature engineering and selection were conducted to analyze how different input features affect the performance of the logistic regression model. Since logistic regression predictions depend on the quality and relevance of the input variables, multiple feature sets were tested to determine which variables contribute most to accurate predictions.

	Feature	Coefficient	Absolute Importance
3	WBC Count	-8.765208	8.765208
6	Platelet Count	-6.174117	6.174117
5	RBC PANEL	-1.093198	1.093198
4	Differential Count	-1.085950	1.085950
0	Age	-1.005092	1.005092
7	PDW	-0.740463	0.740463
1	Sex	-0.113155	0.113155
2	Haemoglobin	-0.038401	0.038401

Fig. 4 Absolute Coefficient of Feature

Feature importance analysis was performed using the coefficients of the logistic regression model. The results indicate that WBC Count and Platelet Count are the most influential predictors for dengue classification, having the highest coefficient magnitudes, which means they are good to use in order to get a high model performance. Other variables such as RBC Panel, Differential Count, and Age also contribute to the prediction but with lower contribution. Features such as PDW, Sex, and Haemoglobin show small influence on the model. These results suggest that blood-related indicators, particularly WBC Count and Platelet Count, play a significant role in predicting dengue cases and those with low coefficients show low correlation resulting for low model's performance.

Feature Selection Experiment				
Features Used	Accuracy	Precision	Recall	F1
Age	67.70%	0.46	0.68	0.55
Platelet, WBC, PDW	98.48%	0.99	0.98	0.98
All features	98.48%	0.99	0.98	0.98

Table. 2. Feature Selection Experiment

Based on the table 2, feature Age as the input feature, resulted in the lowest performance with an accuracy of 67.68%. The confusion matrix shows that the model predicted most samples as class 1, failing to correctly identify class 0 cases. This indicates that Age alone is not a strong indicator for predicting dengue, leading to underfitting and poor classification performance.

The second experiment uses the core medical features such as Platelet Count, WBC Count, and PDW, significantly improved the model's performance. The model achieved an accuracy of 98.48%, with high precision, recall, and F1-score values. These features are directly related to blood test indicators that are commonly associated with dengue infection, allowing the model to better distinguish between dengue and non-dengue cases.

Lastly, the last experiment uses all the features available, it produced the same overall accuracy with the second experiment. Although additional variables are added, they did not significantly improve the model's performance because it has weak correlation to the target variable. This suggests that the core medical features, particularly the Platelet Count, WBC Count, and PDW are the most important predictors required for accurate prediction.

C. Decision Threshold Adjustment

Logistic regression predicts the probability that an observation belongs to a particular class. By default, predictions are classified using a decision threshold of 0.5, where probabilities greater than or equal to 0.5 are classified as class 1 and probabilities below 0.5 are classified as class 0. However, adjusting the threshold can influence the balance between false positives and false negatives, which affects the overall model performance.

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Experiment 3 - Threshold = 0.3
Accuracy: 93.94%
Confusion Matrix:
[[ 52 12]
 [ 0 134]]
Classification Report:
              precision    recall  f1-score   support

     0         1.00        0.81        0.90         64
     1         0.92        1.00        0.96        134

 accuracy          0.94         198
 macro avg         0.96        0.91        0.93         198
 weighted avg         0.94        0.94        0.94         198

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Experiment 3 - Threshold = 0.5
Accuracy: 97.47%
Confusion Matrix:
[[ 59  5]
 [  0 134]]
Classification Report:
              precision    recall  f1-score   support

     0         1.00      0.92      0.96         64
     1         0.96      1.00      0.98        134

   accuracy              0.97         198
  macro avg              0.98      0.96      0.97         198
 weighted avg              0.98      0.97      0.97         198

=====
Experiment 3 - Threshold = 0.7
Accuracy: 99.49%
Confusion Matrix:
[[ 63  1]
 [  0 134]]
Classification Report:
              precision    recall  f1-score   support

     0         1.00      0.98      0.99         64
     1         0.99      1.00      1.00        134

   accuracy              0.99         198
  macro avg              1.00      0.99      0.99         198
 weighted avg              0.99      0.99      0.99         198

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Fig. 5 Decision Threshold Evaluation

In the context of logistic regression, a False Positive occurs when the model predicts that a patient has dengue when they actually do not, while a False Negative occurs when the model predicts that a patient does not have dengue when they actually do.

While observing, when the threshold was set to 0.3, the model produced 12 false positives and only 0 false negative. This means the model classified more cases as dengue, increasing the detection of positive cases but also misclassifying many healthy patients. At the default threshold of 0.5, the model achieved the best balance between the two types of errors, with 5 false positives and 0 false negative. Then, when the threshold was increased to 0.7, the model became more cautious when predicting dengue cases. This reduced the number of false positives to only 1, while still maintaining 0 false negatives.

To summarize the findings, lowering the threshold increases the likelihood of detecting dengue cases but produces more false positives, while increasing the threshold reduces false positives. In medical applications such as dengue prediction, false negatives are more important because missing a dengue case may delay treatment and increase health risks. Therefore, selecting an appropriate threshold is important to balance prediction accuracy and clinical safety.

D. Cross Validation

Cross validation was performed to evaluate the stability and generalization ability of the logistic regression model using WBC Count as the predictor variable. A 5-fold cross validation technique was used, where the dataset was divided into five equal subsets. In each iteration, four subsets were used for training while the remaining subset was used for testing. This process was repeated five times so that each subset served as the testing set once.

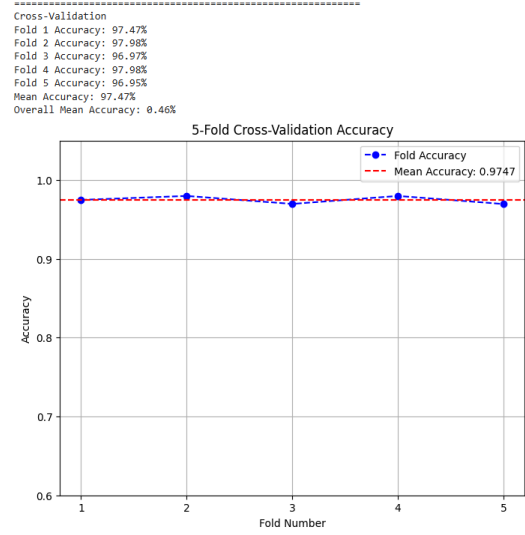


Fig. 6 Cross Validation Output using WBC Count

From the results, the model achieved very similar accuracy across all folds, the average accuracy was 97.47%, which shows that the model performs consistently when different portions of the dataset are used for training and testing. This indicates that the model is stable and not overly dependent on a specific subset of data. Since the model maintained high accuracy across all folds, it suggests that WBC Count is a strong predictor for identifying dengue cases.

IV. PERFORMANCE EVALUATION

The best performing model in this study is the third experiment, which uses all features. Despite having the same output with the second model. By incorporating all available medical indicators, the model can learn more meaningful patterns related to dengue compared to fewer features. Moreover, the model also uses the hyperparameter that is set to $C = 1.0$, which is the default regulation to balance the model complexity and generalization. Moreover the decision threshold is set to 0.5(standard), a 0.7 can also be used but the model might me overfitting. A probability greater than 0.5 indicates that the patient is predicted to be infected with dengue, while a probability lower than 0.5 indicates that the patient is not infected with dengue. The model achieved 97.47% accuracy, demonstrating a strong predictive performance.

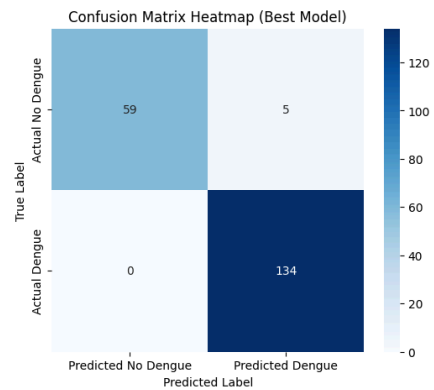


Fig. 6 Confusion matrix of All features

The figure 6, illustrates the performance of the best-performing model in classifying dengue and non-dengue cases. The model correctly predicted 59 cases as No Dengue (true negatives) and 134 cases as Dengue (true positives). There were 5 cases where the model incorrectly predicted Dengue for patients who did not have dengue (false positives). Moreover, the model produced 0 false negatives, meaning that it did not miss any actual dengue cases. This result indicates that the model is highly effective at identifying dengue infections while maintaining a low number of misclassifications. Overall, the confusion matrix supports the model's strong predictive performance, achieving an accuracy of 97.47%, and demonstrates that it is reliable for distinguishing between dengue and non-dengue cases.

Accuracy: 0.9747

Classification Report:				
	precision	recall	f1-score	support
0	1.00	0.92	0.96	64
1	0.96	1.00	0.98	134
accuracy			0.97	198
macro avg	0.98	0.96	0.97	198
weighted avg	0.98	0.97	0.97	198

Fig. 7 Classification Report of All features

To further support the model predictions the figure 7, provides a detailed evaluation of the model's performance using precision, recall, and F1-score. For the No Dengue class (Class 0), the model achieved a precision of 1.00, meaning all predictions labeled as No Dengue were correct. The recall is 0.92, indicating that the model correctly identified 92% of the actual non-dengue cases, while a small portion was misclassified. The F1-score for this class is 0.96, reflecting a strong balance between precision and recall. For the Dengue class (Class 1), the model obtained a precision of 0.96, meaning that 96% of the cases predicted as dengue were actually dengue cases. The recall is 1.00, which indicates that the model successfully identified all actual dengue cases without missing any. The F1-score is 0.98, demonstrating excellent performance in detecting dengue cases.

Overall, the results demonstrate that logistic regression is effective for predicting dengue cases using clinical features. The model using all features achieved high accuracy (97.47%) and strong performance across evaluation metrics, including precision, recall, and F1-score. The confusion matrix shows that the model correctly identified most cases and did not miss any actual dengue cases, which is important for medical diagnosis. In addition, the cross-validation results indicate that the model performs consistently across different data splits. These findings suggest that logistic regression is a reliable and suitable approach for dengue prediction based on medical data.

V. CONCLUSION

This study uses logistic regression to predict dengue infection using clinical and laboratory indicators from the dengue disease dataset. The analysis revealed that WBC Count and Platelet Count are the most important predictors in the dataset because they have the strongest coefficient values and are strongly associated with dengue. These features significantly contributed to the model's ability to distinguish between dengue and non-dengue cases.

Among the experiments conducted, the best-performing model uses all available features with the hyperparameter $C = 1.0$ and the default decision threshold of 0.5. The value $C = 1.0$ provided balanced regularization, allowing the model to learn meaningful relationships without excessive underfitting or overfitting. The decision threshold of 0.5 classifies a case as dengue when the predicted probability is greater than or equal to 0.5. With these settings, the model achieved an accuracy of 97.47% along with high precision, recall, and F1-score values.

Initially, the very high accuracy and evaluation metrics raised some concerns regarding the possibility of model overfitting. However, further analysis of the logistic regression coefficients and feature importance showed that the key predictors, particularly WBC Count and Platelet Count, are strongly correlated with dengue diagnosis and are medically meaningful indicators. This suggests that the high performance is not solely due to overfitting but is likely influenced by the strong predictive power of these clinical variables.

The confusion matrix results further confirmed the model's effectiveness, correctly identifying 134 dengue cases and 59 non-dengue cases, with only 5 false positives and no false negatives. These findings suggest that logistic regression can serve as a useful support tool for dengue prediction using clinical data. By leveraging important medical indicators, such models could assist healthcare professionals in identifying potential dengue cases earlier and improving diagnostic decision-making.

References

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